

Erdős Problem #44

EP-44

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $N \geq 1$ and $A \subset \{1, \dots, N\}$ be a Sidon set. Is it true that, for any $\epsilon > 0$, there exist M and $B \subset \{N+1, \dots, M\}$ (which may depend on N, A, ϵ) such that $A \cup B \subset \{1, \dots, M\}$ is a Sidon set of size at least $(1 - \epsilon)M^{1/2}$?